

National University of Computer and Emerging Sciences
Islamabad Campus

**Web
Programming
(CS4032)**

Course Instructor(s):

Dr. Tajwar Mehmood

**Section(s): (if
applicable)**

Roll No

Course Section

Student Signature

Do not write below this line.

Sessional-I Exam

Attempt all the questions.

Total Time (Hrs): 1

Total Marks: 60

Total Questions: 6

CLO 1. Explain the role and significance of modern web application development technologies, tools, and industry best practices.

Date: Feb 21, 2026

CLO 2. Design and develop dynamic, responsive, and scalable web applications using modern front-end and back-end frameworks.

NOTE: Complete the missing parts (____) to create a simple asked task.

[CLO 2]

Q1: Student Grade Calculator JavaScript Code.

[10 marks]

1. Takes a student's name
2. Takes marks for 3 subject
3. Calculates the average
4. Displays the grade

Fill in the Blanks

Code:

```
<h2>Student Grade Calculator</h2>
Name: <input type="text" id="studentName"><br><br>
Math: <input type="number" id="math"><br><br>
Science: <input type="number" id="science"><br><br>
English: <input type="number" id="english"><br><br>
<button onclick="_____">Calculate</button>
<h3 id="result"></h3>
<script>
function calculateGrade() {
    let name = document._____("____").____;
    let math = Number(document._____("____").____);
    let science = Number(document._____("____").____);
    let english = Number(document._____("____").____);
    let average = (_____) / ____;
    let grade;
```

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```
if (average >= _____) {  
    grade = "A";  
} else if (average >= _____) {  
    grade = "B";  
} else if (average >= _____) {  
    grade = "C";  
} else {  
    grade = "Fail";  
}  
  
document.getElementById("_____").innerHTML = "Student: " + _____ +  
    "<br>Average: " + average.toFixed(2) +  
    "<br>Grade: " + grade;  
}  
</script>
```

Solution

<h2>Student Grade Calculator</h2>

Name: <input type="text" id="studentName">

Math: <input type="number" id="math">

Science: <input type="number" id="science">

English: <input type="number" id="english">

<button onclick="calculateGrade()">Calculate</button>

<h3 id="result"></h3>

<script>

```
function calculateGrade() {  
    let name = document.getElementById("studentName").value;  
    let math = Number(document.getElementById("math").value);  
    let science = Number(document.getElementById("science").value);  
    let english = Number(document.getElementById("english").value);
```

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```
let average = (math + science + english) / 3;
```

```
let grade;
```

```
if (average >= 90) {  
    grade = "A";  
} else if (average >= 70) {  
    grade = "B";  
} else if (average >= 50) {  
    grade = "C";  
} else {  
    grade = "Fail";  
}
```

```
document.getElementById("result").innerHTML =  
    "Student: " + name +  
    "<br>Average: " + average.toFixed(2) +  
    "<br>Grade: " + grade;  
}
```

</script>

[CLO 2]

Q2: Smile Toggle Task using JQuery . Complete the blanks to create a small web page where: [10 marks]

- Clicking the button changes 😞 to 😊
- Clicking again changes it back
- Uses `.text()` and `.click()`

<style>

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```
body {  
  text-align: center;  
  font-family: Arial;  
  margin-top: 100px;  
}
```

```
#face {  
  font-size: 80px;  
}
```

```
button {  
  padding: 10px 20px;  
  font-size: 16px;  
}
```

```
</style>  
</head>  
<body>
```

```
<h2>Smile Toggle</h2>
```

```
<div id="face">😞</div>
```

```
<br>
```

```
<button id="smileBtn">Change Mood</button>
```

```
<script>
```

```
$(_____)._____(function() {
```

```
$("#_____")._____(function() {
```

```
let currentFace = $("#face")._____();
```

```
if (currentFace === "😞") {  
  $("#face")._____("_____");  
} else {  
  $("#face")._____("😞");  
} }); };
```

Solution

Filled blanks in order:

1. document
2. ready
3. #smileBtn
4. click
5. text

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6. text

7. 😊

8. text

[CLO 1]

Q3: Write the output of following script in Asynchronous flow

[2 marks]

```
<script>
function myDisplayer(some) {
  document.getElementById("demo").innerHTML += some + " ";
}
myDisplayer("A");
setTimeout(function() {
  myDisplayer("B");
}, 1000);
myDisplayer("C");
</script>
```

Answer:

Output

A C B

Explanation (Asynchronous flow):

1. myDisplayer("A"); → runs immediately → output: A
2. setTimeout(..., 1000); → scheduled to run after 1 second (asynchronous)
3. myDisplayer("C"); → runs immediately → output: C
4. After 1 second, myDisplayer("B"); runs → output: B

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[CLO 2]

Q4: Match the following images with their code snippet

[10 marks]

This div element has a top padding of 25px, a right and left padding of 50px, and a bottom padding of 75px.

Image 1

div 1

Hooray!

Image 2

div 1

Hooray!

Image 3

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Image 4



Image 5

Code Snippet	Image Number
<pre><style> div { border: 1px solid black; padding: 5px 50px 75px; background-color: lightblue; } </style></pre>	1
<pre><style> .div1 { width: 300px; height: 100px; border: 1px solid blue; box-sizing: border-box; } .div2 { width: 300px; height: 100px; padding: 50px; border: 1px solid red; box-sizing: border-box; } </style> </head> <body> <div class="div1">div 1</div>
 <div class="div2">Hooray!</div></body></pre>	2
<pre><style> .div1 { width: 300px; height: 100px;</pre>	3

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<pre>border: 1px solid blue; } .div2 { width: 300px; height: 100px; padding: 50px; border: 1px solid red; } </style> </head> <body> <div class="div1">div 1</div>
 <div class="div2">Hooray!</div> </body></pre>	
<pre>.flex-container { display: flex; flex-direction: row; }</pre>	4
<pre><style> #example1 { background-color: lightblue; height: 100px; width: 200px; border-start-end-radius: 50px; } </style></pre>	5

[CLO 1]

Q5: Write the code for Resize the browser window to see the effect.

[10 marks]

On screens that are 992px wide or less, the columns will resize from four columns to two columns. On screens that are 600px wide or less, the columns will stack on top of each other instead of next to each other.

Complete the blanks:

```
<head>
<meta name="_____ " content="width=device-width, initial-scale=1">
<style>
* {
box-sizing: _____;
}
```

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```
/* Container for flexboxes */
.container {
  display: _____;
  flex-wrap: wrap;
}

/* Create four equal columns */
.column {
  flex: _____%;
  padding: 20px;
}

/* On screens that are 992px wide or less, go from four columns to two columns */
@media screen and (_____){
  .column {
    flex: _____%;
  }
}

/* On screens that are 600px wide or less, make the columns stack on top of each other instead of next to each other */
@media screen and (_____){
  .container {
    flex-direction: _____;
  }
}
</style>
</head>
<body>
<div class="container">
  <div class="column" style="background-color:#aaa;">
    <h2>Column 1</h2>
    <p>Some text..</p>
  </div>

  <div class="column" style="background-color:#bbb;">
    <h2>Column 2</h2>
    <p>Some text..</p>
  </div>

  <div class="column" style="background-color:#ccc;">
    <h2>Column 3</h2>
    <p>Some text..</p>
  </div>

  <div class="column" style="background-color:#ddd;">
    <h2>Column 4</h2>
    <p>Some text..</p>
  </div>
</div>
```

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```
</body>
async function runSteps() {
  let value = await step1();
  value = await step2(value);
  value = await step3(value);
  myDisplayer(value);
}
```

```
// Call the async function
runSteps();
```

Solution

Filled blanks in order:

1. viewport
2. border-box
3. flex
4. 25
5. max-width: 992px
6. 50
7. max-width: 600px
8. column

How it works (short explanation):

- flex: 25% → 4 columns
- flex: 50% → 2 columns on medium screens
- flex-direction: column → stacked columns on small screens
- viewport → makes layout responsive on mobile devices

[CLO 1]

Q6: Write the following code using same flow with async and await

[10 marks]

```
<body>
<h1>JavaScript Promises</h1>
```

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<h2>Running Functions in Steps</h2>

```
<p id="demo"></p>
<script>
// Function to display something
function myDisplayer(some) {
  document.getElementById("demo").innerHTML = some;
}
// Three functions to run in steps
function step1() {
  return Promise.resolve("A");
}
function step2(value) {
  return Promise.resolve(value + "B");
}
function step3(value) {
  return Promise.resolve(value + "C");
}
// Run the three functions in steps
step1()
.then(function(value) {
  return step2(value);
})
.then(function(value) {
  return step3(value);
})
.then(function(value) {
  myDisplayer(value);
});
</script>
]
```

Answer :

```
<body>
<h1>JavaScript Promises</h1>
<h2>Running Functions in Steps</h2>
```

```
<p id="demo"></p>
```

```
<script>
```

```
// Function to display something
function myDisplayer(some) {
```

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```
document.getElementById("demo").innerHTML = some;  
}
```

```
// Three functions to run in steps
```

```
function step1() {  
    return Promise.resolve("A");  
}
```

```
function step2(value) {  
    return Promise.resolve(value + "B");  
}
```

```
function step3(value) {  
    return Promise.resolve(value + "C");  
}
```

```
// Async function to run steps in sequence
```

```
async function runSteps() {  
    let value = await step1();  
    value = await step2(value);  
    value = await step3(value);  
    myDisplayer(value);  
}
```

```
// Call the async function
```

```
runSteps();
```

```
</script>
```

```
</body>
```